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Editorial Board
Communications in Mathematical Physics
Springer

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

"Existence of a Mass Gap in Lattice Yang-Mills Theory: A Discrete Formulation of the Millennium Problem,"

for consideration for publication in *Communications in Mathematical Physics*.

Summary. The paper proves the existence of a strictly positive mass gap $m(a, \beta) > 0$ in pure $\mathrm{SU}(N)$ Yang-Mills gauge theory on a finite four-dimensional Euclidean lattice. The proof uses Wilson's lattice gauge The proof uses Wilson's lattice gauge theory framework (Wilson 1974), constructs the transfer matrix as a self-adjoint, strictly positive operator on a finite-dimensional Hilbert space, and applies the Perron-Frobenius theorem to guarantee simplicity of the leading eigenvalue—which directly implies the mass gap.

Contributions. The paper provides:

- A rigorous construction of the lattice $\mathrm{SU}(N)$ Yang-Mills partition function as a finite-dimensional integral.
- A proof that the transfer matrix is self-adjoint, strictly positive, and trace-class (following Lüscher 1977).
- A formal proof of the mass gap for all $a > 0$ via Perron-Frobenius.
- Strong-coupling expansion yielding $m \to \infty$ as $g \to \infty$.
- A physical argument, grounded in Planck-scale discreteness, that the lattice formulation is physically fundamental.

Suggested Reviewers.

1. **Prof. Arthur Jaffe** (Harvard) — Co-author of the official Clay Millennium Problem statement for Yang-Mills.
2. **Prof. Martin Lüscher** (CERN/Bern) — Constructed the original transfer matrix for lattice gauge theory.
3. **Prof. Barry Simon** (Caltech) — Leading expert in functional integration and quantum field theory.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

Respectfully submitted,

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